Math 871 Topology I
Problem Set 9

1. Let $p\colon X \longrightarrow Y$ be a continuous map between topological spaces.

   (a) Show that if there is a continuous map $f\colon Y \to X$ such that $p \circ f = \mathrm{id}_Y$, then p is a quotient map.

   (b) Prove that every retraction is a quotient map.

2. Let $X = \mathbb{R}$ with the standard topology and define an equivalence relation on X by

$$x \sim y \iff x - y \in \mathbb{Q}.$$

   Let $Y = X/\sim$ be the quotient of X by $\sim$. Show that Y is not Hausdorff.

   Hint: what is the topology on Y?

3. Consider the unit disk D^2, the subspace of $\mathbb{R}^2$ under the standard topology given by

$$D^2 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leqslant 1\}.$$

   Define an equivalence relation $\sim$ on D^2 by

$$(x_1, y_1) \sim (x_2, y_2) \iff x_1 = x_2.$$

   Show that $D^2/\sim$ is homeomorphic to the interval $[-1, 1]$ under the standard topology.